Project: Worked examples
Title: Bolted joint (simplified VDI 2230)
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1. Scope and method

One bolt clamping members under an axial working load: the preload from the tightening torque with its scatter (or as measured), the stiffness of bolt and members, how much of the working load the bolt sees, the residual clamp after embedding, separation, yield under preload plus working load, the thread engagement needed against stripping, and the fatigue amplitude.

WHAT IT CALCULATES	RELATION
Tensile stress area of the thread	$A_s = (\pi/4)\cdot((d_2 + d_3)/2)^2$, with $d_2 = d - 0.6495\ P$ and $d_3 = d - 1.2269\ P$
Assembly preload from the tightening torque, nut-factor method	$F_M = M_A/(K\ d)$
Assembly preload from the tightening torque, thread and head friction method	$F_M = M_A/(0.16\ P + 0.58\ d_2\ \mu_G + 0.5\ D_{Km}\ \mu_K)$
Assembly preload stated directly (measured) instead of derived	$F_M = F_V$
Thread torsion under preload	$\tau_M = M_G/W_p$ with $M_G = F_M\ (0.16\ P + 0.58\ d_2\ \mu_G)$, $W_p = \pi/16\cdot d_3^3$ at the minor diameter (conservative: the VDI 2230 mean diameter $d_S = (d_2 + d_3)/2$ gives a lower τ_M)
Bolt and member stiffness and the stress area may also be given (measured, FE or a fastener record) instead of derived.	
Minimum preload after the tightening scatter	$F_{M,min} = F_M/\alpha_A$
Bolt resilience, the head, free shank and engaged thread as springs in series	$\delta_S = (0.4\ d + 0.4\ d)/(E_S\ A_N) + l_k/(E_S\ A_s) + 0.5\ d/(E_S\ A_{d3})$
Member stiffness, the clamped parts as two pressure frusta	$k = 0.5774\ \pi\ E\ d\ /\ \ln[((1.155\ t + d_w - d_h)(d_w + d_h))/((1.155\ t + d_w + d_h)(d_w - d_h))]$
Load factor, the share of the working load the bolt sees	$\Phi = n\ k_S/(k_S + k_P)$
Preload lost to embedding (settling of the interfaces)	$F_Z = f_Z\ (k_S\ k_P)/(k_S + k_P)$
Residual clamp at the interface under the working load	$F_{KR} = \max(F_{M,min} - (1 - \Phi)\ F_A - F_Z, 0)$
Maximum bolt load, the whole working load once the joint opens	$F_{S,max} = \max(F_M + \Phi\ F_A, F_A)$
Working equivalent stress in the bolt, tension with half the tightening torsion	$\sigma_{red} = \sqrt{(\sigma_z^2 + 3\ (0.5\ \tau)^2)}$
Fatigue stress amplitude in the thread	$\sigma_a = (F_{S,max} - F_M)/(2\ A_s)$

WHAT IT CALCULATES	RELATION
Thread engagement length needed against stripping	$m_{min} = S\,F_{S,max}/(0.5\,\pi\,d\,\tau_B)$

Basis: Systematic calculation of highly stressed bolted joints — calculation steps R0–R13[1](§5.1–§5.5); Tensile stress area of the thread[2](§9.1.6); Joint stiffness (frusta); preload and torque[3](§8-5, §8-7).

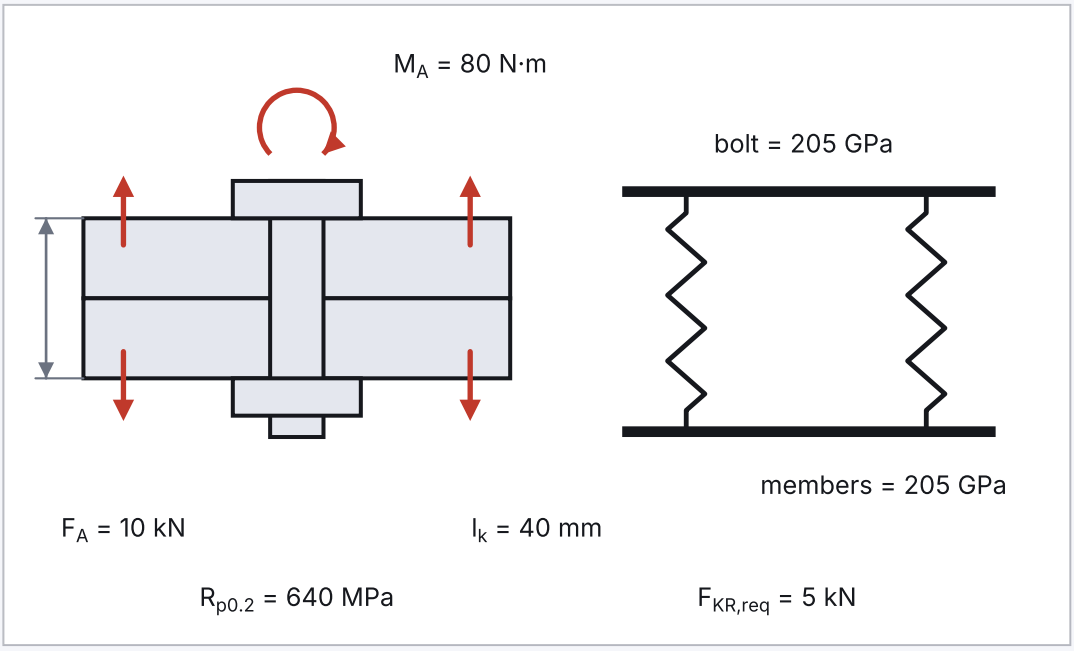


Figure 1 — A preloaded bolt clamping two members, the working load pulling them apart; beside it, bolt and members as two springs in parallel. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
d	Nominal thread diameter	12	mm	drawing: DRG-FLG-02 rev A	3 … 64 mm
P	Thread pitch	1.75	mm	spec: ISO 261 coarse pitch	0.35 … 6 mm
R_{p02}	Bolt 0.2 % proof (yield) strength	640	MPa	spec: ISO 898-1 class 8.8	200 … 1200 MPa
E_S	Bolt Young's modulus	205	GPa	spec: steel	60 … 250 GPa
σ_A	Bolt endurance limit (stress amplitude)	50	MPa	assumption: Rolled-before-heat-treatment M12, VDI 2230 order of magnitude.	10 … 200 MPa
l_k	Clamp length	40	mm	drawing: DRG-FLG-02 rev A	1 … 2000 mm
d_h	Hole diameter	13	mm	drawing: DRG-FLG-02 rev A	1 … 100 mm
d_w	Head / washer bearing diameter	18	mm	spec: ISO 4762 head bearing	2 … 200 mm
E_P	Clamped members' Young's modulus	205	GPa	spec: steel	5 … 250 GPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
n	Load introduction factor (0 = at the clamped faces, 1 = under the head)	0.5		assumption: Load introduced at mid-thickness.	0 … 1
f_Z	Embedding (settling) loss	11	µm	assumption: Three interfaces, machined.	0 … 100 µm
method	Preload method	friction		spec: Assembly procedure AP-7	any
M_A	Tightening torque	80	N·m	spec: Assembly procedure AP-7	0.1 … 10000 N·m
K	Nut factor (nut-factor method); F_M is the maximum preload, so enter the smallest value the joint can have	0.2		assumption: Not used by the friction method.	0.1 … 0.35
μ_G	Thread friction coefficient; F_M is the maximum preload, so enter the smallest value the joint can have	0.12		assumption: Lightly oiled zinc-plated thread.	0.06 … 0.25
μ_K	Head / nut bearing friction coefficient; F_M is the maximum preload, so enter the smallest value the joint can have	0.12		assumption: Lightly oiled head bearing.	0.06 … 0.25
ν_{perm}	Permitted assembly utilisation ν: the equivalent stress at assembly over the proof strength (VDI 2230 R7; 0.9 is the usual target, 1.0 is the limit at which the bolt yields in tightening)	1		default	0.5 … 1
α_A	Tightening scatter factor ($F_{M,max}$ / $F_{M,min}$)	1.6		assumption: Torque wrench, VDI 2230 Table A8 order.	1 … 4
F_A	Axial working load (per bolt)	10	kN	spec: LOAD-CASE-3	0 … 5000 kN
$F_{KR,req}$	Required residual clamp force	5	kN	spec: SPEC-FLG §6 sealing	0 … 5000 kN
τ_B	Nut / internal-thread material shear strength	400	MPa	assumption: Nut material shear strength.	50 … 1200 MPa
m_{eff}	Thread engagement length provided	12	mm	drawing: DRG-FLG-02 rev A	1 … 200 mm
S_{strip}	Safety factor on stripping	1.25		assumption: Stripping safety factor.	1 … 3

3. Assumptions

A1. The joint is concentrically clamped and concentrically loaded; the members behave as two 30° frusta in series. (Eccentric clamping and loading (VDI 2230 full procedure) are outside this sheet.)
A2. Thread torsion is taken at 50 % in the equivalent stress for the working state. (Common VDI 2230 practice (k_τ = 0.5).)
A3. The thread torsion modulus is taken at the minor diameter d_3, $W_p = \pi/16 \cdot d_3^3$, which is conservative: with the VDI 2230 mean diameter d_S the torsion stress is about 14 % lower and v_M about 5 % lower at the defaults. (Stated choice; the equivalent stress at assembly is the more conservative for it.)
A4. F_M is the maximum preload (the one the tightening torque gives with the smallest friction and nut factor); $F_{M,min} = F_M/\alpha_A$ is the minimum after the scatter. (VDI 2230 R2: the maximum preload comes from the minimum friction coefficients.)
A5. The working load is a pulsating 0…F_A load; the fatigue amplitude is $\Phi\, F_A / (2\, A_s)$. (Compare with the bolt's endurance limit as entered.)
A6. Thread stripping uses a shear area of half the pitch per unit length on the nominal diameter. (A conservative simplification of ISO 898-2 / VDI 2230 §5.5.5.)
A7. σ_A (Bolt endurance limit (stress amplitude)) is taken as 50 MPa. (Rolled-before-heat-treatment M12, VDI 2230 order of magnitude.)
A8. n (Load introduction factor (0 = at the clamped faces, 1 = under the head)) is taken as 0.5. (Load introduced at mid-thickness.)
A9. f_Z (Embedding (settling) loss) is taken as 11 µm. (Three interfaces, machined.)
A10. K (Nut factor (nut-factor method); F_M is the maximum preload, so enter the smallest value the joint can have) is taken as 0.2. (Not used by the friction method.)
A11. μ_G (Thread friction coefficient; F_M is the maximum preload, so enter the smallest value the joint can have) is taken as 0.12. (Lightly oiled zinc-plated thread.)
A12. μ_K (Head / nut bearing friction coefficient; F_M is the maximum preload, so enter the smallest value the joint can have) is taken as 0.12. (Lightly oiled head bearing.)
A13. α_A (Tightening scatter factor ($F_{M,max}$ / $F_{M,min}$)) is taken as 1.6. (Torque wrench, VDI 2230 Table A8 order.)
A14. τ_B (Nut / internal-thread material shear strength) is taken as 400 MPa. (Nut material shear strength.)
A15. S_{strip} (Safety factor on stripping) is taken as 1.25. (Stripping safety factor.)

4. Working

[24] $d_2 = d - 0.6495 * P$ ²(§9.1.6)
$= 12\ mm - 0.6495 \times 1.75\ mm$
$= 10.863\ mm$
Pitch diameter

[25] $d_3 = d - 1.2269 * P$ ²(§9.1.6)
$= 12\ mm - 1.2269 \times 1.75\ mm$
$= 9.8529\ mm$
Minor diameter

[26] $A_s = pi / 4 * ((d_2 + d_3) / 2)^2$ ²(§9.1.6)
$= \pi / 4 \times ((10.863\ mm + 9.8529\ mm) / 2)^2$
$= 84.266\ mm^2$
Tensile stress area

[27] $A_N = pi / 4 * d^2$
$= \pi / 4 \times (12\ mm)^2$
$= 113.1\ mm^2$
Nominal shank area

[28] $A_{d3} = pi / 4 * d_3^2$
$= \pi / 4 \times (9.8529\ mm)^2$
$= 76.247\ mm^2$
Minor-diameter area

[29] $D_{Km} = (d_w + d_h) / 2$
$= (18\ mm + 13\ mm) / 2$
$= 15.5\ mm$
Mean bearing diameter under the head
[30] $F_M = M_A / (0.16 * P + 0.58 * d_2 * \mu_G + 0.5 * D_{Km} * \mu_K)$ [1](§5.4.3)
$= 80\ N·m / (0.16 \times 1.75\ mm + 0.58 \times 10.863\ mm \times 0.12 + 0.5 \times 15.5\ mm \times 0.12)$
$= 40.69\ kN$
Assembly preload (friction)
[31] $F_{M,min} = F_M / \alpha_A$ [1](R2)
$= 40.69\ kN / 1.6$
$= 25.431\ kN$
Minimum preload after scatter
[32] $\sigma_M = F_M / A_s$
$= 40.69\ kN / 84.266\ mm^2$
$= 482.87\ MPa$
Preload stress
[33] $M_G = F_M * (0.16 * P + 0.58 * d_2 * \mu_G)$ [1](§5.4.3)
$= 40.69\ kN \times (0.16 \times 1.75\ mm + 0.58 \times 10.863\ mm \times 0.12)$
$= 42.158\ N·m$
Thread torque under preload
[34] $W_p = pi / 16 * d_3^3$
$= \pi / 16 \times (9.8529\ mm)^3$
$= 187.81\ mm^3$
Polar section modulus at the minor diameter
[35] $\tau_M = M_G / W_p$
$= 42.158\ N·m / 187.81\ mm^3$
$= 224.47\ MPa$
Thread torsional stress under preload
[36] $\sigma_{red,M} = sqrt(\sigma_M^2 + 3 * \tau_M^2)$ [1](R8)
$= sqrt((482.87\ MPa)^2 + 3 \times (224.47\ MPa)^2)$
$= 619.94\ MPa$
Equivalent stress at assembly (full torsion)
[37] $\nu_M = \sigma_{red,M} / R_{p02}$
$= 619.94\ MPa / 640\ MPa$
$= 0.96866$
Utilisation of proof strength at assembly
[38] $u_{assembly} = \nu_M / \nu_{perm}$
$= 0.96866 / 1$
$= 0.96866$
Assembly utilisation over the permitted ν
[39] $\delta_S = 0.4 * d / (E_S * A_N) + l_k / (E_S * A_s) + 0.5 * d / (E_S * A_{d3}) + 0.4 * d / (E_S * A_N)$ [1](§5.4.1)
$= 0.4 \times 12\ mm / (205\ GPa \times 113.1\ mm^2) + 40\ mm / (205\ GPa \times 84.266\ mm^2) + 0.5 \times 12\ mm / (205\ G$
$76.247\ mm^2) + 0.4 \times 12\ mm / (205\ GPa \times 113.1\ mm^2)$
$= 0.0000031135\ m/kN$
Bolt resilience (head, shank, engaged thread, nut)
[40] $k_S = 1 / \delta_S$
$= 1 / 0.0000031135\ m/kN$
$= 321190\ kN/m$
Bolt stiffness
[41] $t_f = l_k / 2$
$= 40\ mm / 2$
$= 20\ mm$
Frustum height (two equal frusta)

[42] $k_f = 0.5774 * pi * E_P * d / ln(((1.155 * t_f + d_w - d_h) * (d_w + d_h)) / ((1.155 * t_f + d_w + (d_w - d_h)))$ [3](eq. 8-20)

$= 0.5774 \times \pi \times 205\ GPa \times 12\ mm / ln((1.155 \times 20\ mm + 18\ mm - 13\ mm) \times (18\ mm + 13\ mm) / ((1.1 mm + 18\ mm + 13\ mm) \times (18\ mm - 13\ mm)))$

$= 3815600\ kN/m$

Stiffness of one frustum

[43] $k_P = k_f / 2$ [3](eq. 8-21)

$= 3815600\ kN/m / 2$

$= 1907800\ kN/m$

Clamped members' stiffness (two frusta in series)

[44] $\Phi = n * k_S / (k_S + k_P)$ [1](R3)

$= 0.5 \times 321190\ kN/m / (321190\ kN/m + 1907800\ kN/m)$

$= 0.072047$

Load factor

[45] $F_Z = f_Z * (k_S * k_P) / (k_S + k_P)$ [1](R4)

$= 11\ \mu m \times 321190\ kN/m \times 1907800\ kN/m / (321190\ kN/m + 1907800\ kN/m)$

$= 3.024\ kN$

Preload loss from embedding

[46] $F_{SA} = \Phi * F_A$

$= 0.072047 \times 10\ kN$

$= 0.72047\ kN$

Additional bolt load from the working load

[47] $F_{PA} = (1 - \Phi) * F_A$

$= (1 - 0.072047) \times 10\ kN$

$= 9.2795\ kN$

Load relieved from the members

[48] $F_{KR} = max(F_{M,min} - F_{PA} - F_Z, 0\ kN)$ [1](R7)

$= max(25.431\ kN - 9.2795\ kN - 3.024\ kN, 0\ kN)$

$= 13.128\ kN$

Residual clamp force (minimum; zero once the joint opens)

[49] $F_{S,max} = max(F_M + F_{SA}, F_A)$

$= max(40.69\ kN + 0.72047\ kN, 10\ kN)$

$= 41.41\ kN$

Maximum bolt load (maximum preload + additional load; the whole working load once the joint opens)

[50] $\sigma_{z,max} = F_{S,max} / A_s$

$= 41.41\ kN / 84.266\ mm^2$

$= 491.42\ MPa$

Maximum bolt tensile stress

[51] $\sigma_{red,B} = sqrt(\sigma_{z,max}^2 + 3 * (0.5 * \tau_M)^2)$ [1](R8)

$= sqrt((491.42\ MPa)^2 + 3 \times (0.5 \times 224.47\ MPa)^2)$

$= 528.47\ MPa$

Equivalent stress in the working state (50 % torsion)

[52] $u_{yield} = \sigma_{red,B} / R_{p02}$

$= 528.47\ MPa / 640\ MPa$

$= 0.82574$

Yield utilisation in the working state

[53] $\sigma_a = (F_{S,max} - F_M) / (2 * A_s)$ [1](R9)

$= (41.41\ kN - 40.69\ kN) / (2 \times 84.266\ mm^2)$

$= 4.2749\ MPa$

Fatigue stress amplitude (pulsating 0…F_A load)

[54] $u_{fatigue} = \sigma_a / \sigma_A$

$= 4.2749\ MPa / 50\ MPa$

$= 0.085499$

Fatigue utilisation

[55] $m_{min} = S_{strip} * F_{S,max} / (0.5 * pi * d * \tau_B)$ [1](§5.5.5)
$= 1.25 \times 41.41\ kN / (0.5 \times \pi \times 12\ mm \times 400\ MPa)$
$= 6.8653\ mm$
Thread engagement needed against stripping
[56] $u_{strip} = m_{min} / m_{eff}$
$= 6.8653\ mm / 12\ mm$
$= 0.57211$
Engagement utilisation
[57] $F_{A,sep} = (F_{M,min} - F_Z) / (1 - \Phi)$ [1](R7)
$= (25.431\ kN - 3.024\ kN) / (1 - 0.072047)$
$= 24.147\ kN$
Working load at which the members separate (minimum preload, after embedding)
[58] $F_{proof} = R_{p02} * A_s$
$= 640\ MPa \times 84.266\ mm^2$
$= 53.93\ kN$
Bolt load at the proof (yield) strength
[59] $n_0 = F_{A,sep} / F_A$
$= 24.147\ kN / 10\ kN$
$= 2.4147$
Load factor against separation ($F_{A,sep} / F_A$)
[60] $F_{A,yield} = \min((F_{proof} - F_M) / \Phi, F_{proof})$
$= \min((53.93\ kN - 40.69\ kN) / 0.072047, 53.93\ kN)$
$= 53.93\ kN$
Working load at which the bolt reaches its proof load (separation included)
[61] $n_L = F_{A,yield} / F_A$
$= 53.93\ kN / 10\ kN$
$= 5.393$
Load factor against axial yield of the bolt, torsion excluded
[62] $u_{clamp} = F_{KR,req} / F_{KR}$
$= 5\ kN / 13.128\ kN$
$= 0.38087$
Required over available residual clamp

5. Results

SYMBOL	VALUE	DESCRIPTION
n_0	2.4147	Load factor against separation
n_L	5.393	Load factor against axial yield
A_s	84.266 mm²	Tensile stress area
F_M	40.69 kN	Assembly preload (maximum)
$F_{M,min}$	25.431 kN	Minimum preload
σ_M	482.87 MPa	Preload stress
ν_M	0.96866	Assembly utilisation of proof strength
$u_{assembly}$	0.96866	Assembly utilisation over the permitted ν
k_S	321190 kN/m	Bolt stiffness
k_P	1907800 kN/m	Members' stiffness
Φ	0.072047	Load factor
F_Z	3.024 kN	Embedding loss

SYMBOL	VALUE	DESCRIPTION
F_{SA}	0.72047 kN	Additional bolt load
F_{KR}	13.128 kN	Residual clamp
$F_{A,sep}$	24.147 kN	Separation load
$F_{S,max}$	41.41 kN	Maximum bolt load
$\sigma_{z,max}$	491.42 MPa	Maximum bolt tensile stress
$\sigma_{red,B}$	528.47 MPa	Working equivalent stress
σ_a	4.2749 MPa	Fatigue amplitude
m_{min}	6.8653 mm	Engagement needed

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Equivalent stress at assembly within the permitted ν of the proof strength	$u_{assembly}$ = 0.96866	$u_{assembly} \le 1$	0.969	[1](R7)
PASS	Yield in the working state	u_{yield} = 0.82574	$u_{yield} \le 1$	0.826	[1](R8)
PASS	Thread stripping	u_{strip} = 0.57211	$u_{strip} \le 1$	0.572	[1](§5.5.5)
PASS	Residual clamp against requirement	u_{clamp} = 0.38087	$u_{clamp} \le 1$	0.381	[1](R7)
PASS	Fatigue amplitude	$u_{fatigue}$ = 0.085499	$u_{fatigue} \le 1$	0.0855	[1](R9)
PASS	No separation	13.128 kN	> 0 kN	0	[1](R7)

7. Applicability

WARNING — input.nu_perm: ν_{perm} (Permitted assembly utilisation ν: the equivalent stress at assembly over the proof strength (VDI 2230 R7; 0.9 is the usual target, 1.0 is the limit at which the bolt yields in tightening)) was not entered; the template default 1 is used.
WARNING — preload-yield: M_A: the preload alone takes the bolt above 90 % of its proof strength.

8. Verification

Template mech.bolted-joint@1.2.1 Content hash df65ae01a912bd25 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Verein Deutscher Ingenieure. Systematic Calculation of Highly Stressed Bolted Joints: Joints With One Cylindrical Bolt. VDI 2230 Part 1:2015. Beuth Verlag; 2015.
2. International Organization for Standardization. Mechanical Properties of Fasteners Made of Carbon Steel and Alloy Steel—Part 1: Bolts, Screws and Studs With Specified Property Classes—Coarse Thread and Fine Pitch Thread. ISO 898-1:2013. ISO; 2013.
3. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.